Q.1 On the given graph, if we perform DFS starting from node A. Which of the following could be a valid DFS traversal order? Graph (Adjacency List): A → B, C B → (no outgoing edges) C → (no outgoing edges)

A. A, B, C

B. C, B, A

C. C, A, B

D. B, C, A

Q.2 _____ condition is sufficient to detect cycle in a directed graph.

A. There is an edge from currently being visited node to an already visited node

B. There is an edge from currently being visited node to an ancestor of currently visited node in DFS forest

C. Every node is seen twice in DFS

D. Every edge is seen twice in DFS

Q.3 What will be the time complexity of the iterative depth first search traversal code , where V = No. of Vertices, E = No. of Edges ?

A. O(V+E)

B. O(V)

C. O(E)

D. O(V*E)

Q.4 On the given graph, if DFS starts from A and visits C first, what is the traversal order? Graph (Adjacency List): A → B, C B → C C → (no outgoing edges)

A. B, C, A

B. A, B, C

C. B, A, C

D. A, C, B

Q.5 If DFS starts from A on the given graph, which node will NOT be visited? Graph (Adjacency List): A → B, C B → C D → A

A. B

B. D

C. A

D. C
